

OOPs assignment 3

Submitted by: Mehma Kaur Chadha

Roll Number 1024150080

Batch:2o13

Q1 Write a program to implement (a) pointer to an object (b) this pointer.
Practice both '.' (dot operator) and '→' (arrow operator).

Code:

```
main.cpp
1  #include <iostream>
2  using namespace std;
3
4  class Demo {
5      int x;
6  public:
7      void set(int x) {
8          this->x = x;
9      }
10     void show() {
11         cout << x << endl;
12     }
13 };
14
15 int main() {
16     Demo d1;
17     Demo *ptr;
18
19     d1.set(10);
20     d1.show();
21
22     ptr = &d1;
23     ptr->set(20);
24     ptr->show();
25
26     return 0;
27 }
```

Output:

```
10
20

...Program finished with exit code 0
Press ENTER to exit console.
```

Q2 Write a program to swap private values of two classes using a friend function.

```
main.cpp
1  #include <iostream>
2  using namespace std;
3
4  class B;
5
6  class A {
7      int x;
8  public:
9      void set(int a) { x = a; }
10     void show() { cout << x << endl; }
11     friend void swap(A&, B&);
12 };
13
14 class B {
15     int y;
16 public:
17     void set(int b) { y = b; }
18     void show() { cout << y << endl; }
19     friend void swap(A&, B&);
20 };
21
22 void swap(A &a, B &b) {
23     int temp = a.x;
24     a.x = b.y;
25     b.y = temp;
26 }
27
28 int main() {
29     A a;
30     B b;
```

```

13
14 class B {
15     int y;
16 public:
17     void set(int b) { y = b; }
18     void show() { cout << y << endl; }
19     friend void swap(A&, B&);
20 };
21
22 void swap(A &a, B &b) {
23     int temp = a.x;
24     a.x = b.y;
25     b.y = temp;
26 }
27
28 int main() {
29     A a;
30     B b;
31
32     a.set(5);
33     b.set(10);
34
35     swap(a, b);
36
37     a.show();
38     b.show();
39
40     return 0;
41 }

```

input

```

10
5

...Program finished with exit code 0
Press ENTER to exit console.

```

Q3 Write a program to add data objects of two different classes using friend functions.

main.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  class B;
5
6  class A {
7      int x;
8  public:
9      void set(int a) { x = a; }
10     friend int add(A, B);
11 };
12
13 class B {
14     int y;
15 public:
16     void set(int b) { y = b; }
17     friend int add(A, B);
18 };
19
20 int add(A a, B b) {
21     return a.x + b.y;
22 }
23
24 int main() {
25     A a;
26     B b;
27
28     a.set(3);
29     b.set(7);
30 }
```

```
17     friend int add(A, B);
18 };
19
20 int add(A a, B b) {
21     return a.x + b.y;
22 }
23
24 int main() {
25     A a;
26     B b;
27
28     a.set(3);
29     b.set(7);
30
31     cout << add(a, b);
32
33     return 0;
34 }
```

input

10

...Program finished with exit code 0
Press ENTER to exit console.

Q4 Write a program to demonstrate the working of friend class.

```

n.cpp
1  #include <iostream>
2  using namespace std;
3
4  class A {
5      int x;
6  public:
7      void set(int a) { x = a; }
8      friend class B;
9  };
10
11 class B {
12 public:
13     void show(A a) {
14         cout << a.x;
15     }
16 };
17
18 int main() {
19     A a;
20     B b;
21
22     a.set(100);
23     b.show(a);
24
25     return 0;

```

```

100
...Program finished with exit code 0
Press ENTER to exit console.

```

Q5 Write a program using Array of Objects to display area of multiple rectangles.

ain.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  class Rectangle {
5      int l, b;
6  public:
7      void set(int x, int y) {
8          l = x;
9          b = y;
10     }
11     int area() {
12         return l * b;
13     }
14 };
15
16 int main() {
17     Rectangle r[3];
18
19     for(int i = 0; i < 3; i++) {
20         int l, b;
21         cin >> l >> b;
22         r[i].set(l, b);
23     }
24
25     for(int i = 0; i < 3; i++) {
26         cout << r[i].area() << endl;
27     }
28
29     return 0;
30 }
```

input

```
1
2
3
4
5
6
2
12
30
```

...Program finished with exit code 0
Press ENTER to exit console.

